RANS PART 2: REYNOLDS-AVERAGED MOMENTUM EQUATION (X-COMPONENT)

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Fluid Dynamics and IBM Theory • Module 5

Adapted from a lecture by Steven L. Brunton.

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WHERE WE ARE (FROM PART 1)

We start from incompressible Navier–Stokes (nondimensional):

$$\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\nabla p + \frac{1}{\mathrm{Re}}\Delta \boldsymbol{u},$$
$$\nabla \cdot \boldsymbol{u} = 0.$$

Reynolds decomposition (time average):

$$\boldsymbol{u}(\boldsymbol{x}, t) = \overline{\boldsymbol{u}}(\boldsymbol{x}) + \boldsymbol{u}'(\boldsymbol{x}, t), \qquad \overline{\boldsymbol{u}'} = 0.$$

Result from Part 1:

$$\nabla \cdot \overline{\boldsymbol{u}} = 0, \qquad \nabla \cdot \boldsymbol{u}' = 0.$$

OTHER IDENTITIES FROM PART 1

Recall we are taking averages in time. We know

$$\begin{aligned}
\overline{u+v} &= \overline{u} + \overline{v}, \\
\overline{\overline{u}} &= \overline{u}, \\
\overline{\overline{u} \cdot v} &= \overline{u} \cdot \overline{v}, \\
\overline{\frac{\partial u}{\partial x}} &= \frac{\partial \overline{u}}{\partial x}
\end{aligned}$$

In general,

$$\begin{aligned}
\overline{u \cdot v} &\neq \overline{u} \cdot \overline{v}, \\
\overline{u'^2} &\neq 0.
\end{aligned}$$

MOTIVATION: WHY AVERAGE THE MOMENTUM EQUATION?

Engineers often want **mean** quantities:

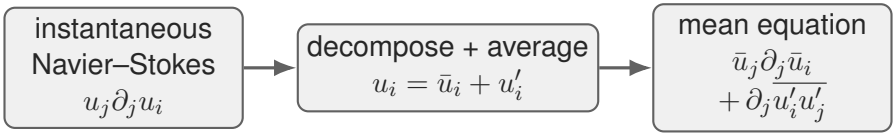
- mean drag / mean lift,
- mean separation behavior,
- bulk transport trends,

without resolving every turbulent eddy in space-time.

Plan today

Plug $\boldsymbol{u} = \overline{\boldsymbol{u}} + \boldsymbol{u}'$ (and $p = \overline{p} + p'$) into momentum, then time-average to get an equation for $\overline{\boldsymbol{u}}$.

WHAT THE AVERAGING OPERATION CHANGES



Schematic: instructor's course notes. Index notation: u_i ($i = 1, 2, 3$) are the components (u, v, w), $\partial_j = \partial/\partial x_j$, and a repeated index is summed.

The nonlinear term does not simply average away

The correlation $R_{ij} = \overline{u_i' u_j'}$ is new information. An equation for the mean velocity alone does not determine it: this is the RANS closure problem.

WE WILL DERIVE ONLY THE X-MOMENTUM EQUATION

Write $\boldsymbol{u} = (u, v, w)^T$. The x-component of momentum is:

$$\frac{\partial u}{\partial t} + u\,\frac{\partial u}{\partial x} + v\,\frac{\partial u}{\partial y} + w\,\frac{\partial u}{\partial z} = -\frac{\partial p}{\partial x} + \frac{1}{\mathrm{Re}}\Delta u.$$

We substitute:

$$u = \overline{u} + u', \quad v = \overline{v} + v', \quad w = \overline{w} + w', \quad p = \overline{p} + p'.$$

AVERAGING IDENTITIES WE WILL USE REPEATEDLY

Time average operator $\overline{\cdot}$ is linear and (under mild regularity) commutes with spatial derivatives:

$$\overline{a+b} = \overline{a} + \overline{b}, \qquad \overline{\frac{\partial}{\partial x} a} = \frac{\partial}{\partial x} \overline{a}.$$

Key facts:

$$\overline{u'} = 0, \qquad \overline{p'} = 0, \qquad \overline{\Delta u'} = \Delta \overline{u'} = 0.$$

Critical warning (the source of Reynolds stresses)

In general,

$$\overline{ab} \neq \overline{a}\,\overline{b}.$$

So averages of *products of fluctuations* need not vanish.

STEP 1: SUBSTITUTE DECOMPOSITION INTO X-MOMENTUM

Start with

$$\frac{\partial u}{\partial t} + u\,\frac{\partial u}{\partial x} + v\,\frac{\partial u}{\partial y} + w\,\frac{\partial u}{\partial z} = -\frac{\partial p}{\partial x} + \frac{1}{\mathrm{Re}}\Delta u.$$

Substitute $u = \overline{u} + u'$, $v = \overline{v} + v'$, $w = \overline{w} + w'$, $p = \overline{p} + p'$. Then

$$\frac{\partial}{\partial t}(\overline{u} + u') + (\overline{u} + u')\frac{\partial}{\partial x}(\overline{u} + u') + (\overline{v} + v')\frac{\partial}{\partial y}(\overline{u} + u') + (\overline{w} + w')\frac{\partial}{\partial z}(\overline{u} + u') = -\frac{\partial}{\partial x}(\overline{p} + p') + \frac{1}{\mathrm{Re}}\Delta(\overline{u} + u').$$

STEP 1A: EXPAND THE CONVECTION TERM (X-PART SHOWN)

The first convection piece expands like multiplying polynomials:

$$(\overline{u}+u')\frac{\partial}{\partial x}(\overline{u}+u') = \overline{u}\,\frac{\partial \overline{u}}{\partial x} + \overline{u}\,\frac{\partial u'}{\partial x} + u'\,\frac{\partial \overline{u}}{\partial x} + u'\,\frac{\partial u'}{\partial x}.$$

Similarly,

$$(\overline{v}+v')\frac{\partial}{\partial y}(\overline{u}+u') = \overline{v}\,\frac{\partial \overline{u}}{\partial y} + \overline{v}\,\frac{\partial u'}{\partial y} + v'\,\frac{\partial \overline{u}}{\partial y} + v'\,\frac{\partial u'}{\partial y},$$

and likewise for the w-term.

STEP 2: TIME-AVERAGE THE ENTIRE EQUATION

Time-average term-by-term.

Time derivative:

$$\overline{\frac{\partial}{\partial t}(\overline{u}+u')} = \frac{\partial}{\partial t}\overline{u} + \frac{\partial}{\partial t}\overline{u'} = \frac{\partial}{\partial t}\overline{u}.$$

For a statistically stationary mean, $\frac{\partial}{\partial t}\overline{u} = 0$. Otherwise this mean-acceleration term must remain.

Pressure and viscosity:

$$\overline{-\frac{\partial}{\partial x}(\overline{p}+p')} = -\frac{\partial}{\partial x}\overline{p}, \qquad \overline{\Delta(\overline{u}+u')} = \Delta\overline{u}.$$

Mixed mean–fluctuation products die: e.g.

$$\overline{\overline{u}\,\frac{\partial u'}{\partial x}} = \overline{u}\,\overline{\frac{\partial u'}{\partial x}} = 0, \quad \overline{u'\,\frac{\partial \overline{u}}{\partial x}} = \overline{u'}\,\frac{\partial \overline{u}}{\partial x} = 0.$$

What survives are *mean–mean* terms and *fluctuation–fluctuation* terms.

STEP 2 RESULT: AVERAGED X-MOMENTUM (STATIONARY MEAN)

Here $\partial_x = \partial/\partial x$ etc.; subscripts are reserved for components.
After averaging, the surviving terms are:

$$\overline{u}\,\partial_x\overline{u} + \overline{v}\,\partial_y\overline{u} + \overline{w}\,\partial_z\overline{u} \;+\; \overline{u'\,\partial_x u'} + \overline{v'\,\partial_y u'} + \overline{w'\,\partial_z u'} = -\partial_x\overline{p} + \frac{1}{\mathrm{Re}}\Delta\overline{u}.$$

Interpretation

Everything is now in terms of mean fields *except* the three correlation terms $\overline{u'\,\partial_x u'}$, $\overline{v'\,\partial_y u'}$, $\overline{w'\,\partial_z u'}$. These encode how fluctuations force the mean.

THE PROBLEMATIC SURVIVORS: WHY THEY DON'T VANISH

Even though $\overline{u'} = 0$, products of fluctuations need not average to zero:

$$\overline{(u')^2} \neq 0, \qquad \overline{u'v'} \neq 0, \qquad \overline{u'w'} \neq 0.$$

These are (components of) the **Reynolds stresses**. They will become **spatial derivatives of** $\overline{(u')^2}$, $\overline{u'v'}$, $\overline{u'w'}$.
This is where closure enters: we need a model for these correlations.

STEP 3: THE STREAMWISE PRODUCT-RULE IDENTITY

Use the product rule:

$$\partial_x\big((u')^2\big) = 2u'\,\partial_x u' \quad \Rightarrow \quad u'\,\partial_x u' = \frac{1}{2}\,\partial_x\big((u')^2\big).$$

Therefore,

$$\overline{u'\,\partial_x u'} = \frac{1}{2}\,\frac{\partial}{\partial x}\overline{(u')^2}.$$

This half-factor is correct for this term in isolation. In the sum of all three convective terms, continuity supplies the complementary half, as the next two slides show explicitly.

STEP 3: PRODUCT-RULE SETUP FOR THE CROSS TERMS

Apply the product rule to $u'v'$ and $u'w'$:

$$\frac{\partial}{\partial y}(u'v') = v'\,\partial_y u' + u'\,\partial_y v', \qquad \frac{\partial}{\partial z}(u'w') = w'\,\partial_z u' + u'\,\partial_z w'.$$

Rearrange:

$$v'\,\partial_y u' = \frac{\partial}{\partial y}(u'v') - u'\,\partial_y v', \qquad w'\,\partial_z u' = \frac{\partial}{\partial z}(u'w') - u'\,\partial_z w'.$$

Now time-average:

$$\overline{v'\,\partial_y u'} = \frac{\partial}{\partial y}\overline{u'v'} - \overline{u'\,\partial_y v'}, \quad \overline{w'\,\partial_z u'} = \frac{\partial}{\partial z}\overline{u'w'} - \overline{u'\,\partial_z w'}.$$

These identities expose derivatives of the correlations, but they also leave two terms involving derivatives of v' and w'. The complete vector product rule organizes all three directions at once.

STEP 3: PACKAGE THE FULL CONVECTIVE SUM

Apply $a_j \partial_j b = \partial_j(a_j b) - b\,\partial_j a_j$ with $a_j = u'_j$ and $b = u'$:

$$u'\,\partial_x u' + v'\,\partial_y u' + w'\,\partial_z u' = \frac{\partial (u')^2}{\partial x} + \frac{\partial (u'v')}{\partial y} + \frac{\partial (u'w')}{\partial z} - u'\left(\partial_x u' + \partial_y v' + \partial_z w'\right).$$

Why the half-factor disappears

As on the streamwise slide: the product-rule correction supplies the second copy of $u'\,\partial_x u'$, so the first term is $\partial_x(u'^2)$, not $\frac{1}{2}\partial_x(u'^2)$.

In detail: $\partial_x(u'^2) = 2u'\partial_x u'$ contains one copy too many; the extra copy is cancelled by the $-u'\partial_x u'$ inside $-u'(\nabla \cdot \boldsymbol{u}')$. That whole bracket vanishes only because all three of its pieces are kept together, which is why the cross terms must be treated in the same step.

THE CANCELLATION: USE $\nabla \cdot \boldsymbol{u}' = 0$

The correction in the packaged identity is

$$\overline{u'\,\partial_x u'} + \overline{u'\,\partial_y v'} + \overline{u'\,\partial_z w'}.$$

Factor out u':

$$u'\,\partial_x u' + u'\,\partial_y v' + u'\,\partial_z w' = u'\,(\partial_x u' + \partial_y v' + \partial_z w') = u'\,(\nabla \cdot \boldsymbol{u}').$$

But from Part 1, $\nabla \cdot \boldsymbol{u}' = 0$, hence

$$\overline{u'\,\partial_x u'} + \overline{u'\,\partial_y v'} + \overline{u'\,\partial_z w'} = 0.$$

The correction vanishes, so no factor of $\frac{1}{2}$ is missing (see the streamwise slide).

STEP 3 RESULT: THE REYNOLDS-STRESS-GRADIENT FORM

Using the identities and cancellations, the fluctuation terms become:

$$\overline{u'\,\partial_x u'} + \overline{v'\,\partial_y u'} + \overline{w'\,\partial_z u'} = \frac{\partial}{\partial x}\overline{(u')^2} + \frac{\partial}{\partial y}\overline{u'v'} + \frac{\partial}{\partial z}\overline{u'w'}.$$

Equivalently, in index notation,

$$\overline{u_j'\partial_j u_i'} = \partial_j \overline{u_i' u_j'} - \overline{u_i'\partial_j u_j'} = \partial_j R_{ij},$$

because $\partial_j u_j' = 0$.

FINAL X-COMPONENT RANS EQUATION (STATIONARY MEAN)

The Reynolds-averaged x-momentum equation can be written as:

$$\overline{u}\,\partial_x\overline{u} + \overline{v}\,\partial_y\overline{u} + \overline{w}\,\partial_z\overline{u} = -\partial_x\overline{p} + \frac{1}{\mathrm{Re}}\Delta\overline{u} - \Big(\frac{\partial}{\partial x}\overline{(u')^2} + \frac{\partial}{\partial y}\overline{u'v'} + \frac{\partial}{\partial z}\overline{u'w'}\Big).$$

Key interpretation

The mean flow is driven by pressure + viscosity *and* the divergence of Reynolds stresses.

THE GENERAL UNSTEADY RANS EQUATION

If the averaging operation produces a time-dependent mean, retain its time derivative:

$$\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{\partial \bar{p}}{\partial x_i} + \frac{1}{\mathrm{Re}} \Delta \bar{u}_i - \frac{\partial R_{ij}}{\partial x_j}, \qquad \frac{\partial \bar{u}_j}{\partial x_j} = 0.$$

Stationary specialization

The preceding derivation set $\partial_t \bar{u}_i = 0$ only after explicitly assuming statistically stationary mean fields. Reynolds averaging itself does not require this term to vanish.

VECTOR/TENSOR FORM (WHAT THIS LOOKS LIKE IN COMPACT NOTATION)

Define the velocity-correlation tensor:

$$\boldsymbol{R} := \overline{\boldsymbol{u}'\,\boldsymbol{u}'^{T}} = \begin{bmatrix} \overline{u'u'} & \overline{u'v'} & \overline{u'w'} \\ \overline{v'u'} & \overline{v'v'} & \overline{v'w'} \\ \overline{w'u'} & \overline{w'v'} & \overline{w'w'} \end{bmatrix}.$$

Then the stationary mean momentum equation is commonly written:

$$(\overline{\boldsymbol{u}} \cdot \nabla)\overline{\boldsymbol{u}} = -\nabla \overline{p} + \frac{1}{\mathrm{Re}} \Delta \overline{\boldsymbol{u}} - \nabla \cdot \boldsymbol{R}.$$

Along with mean incompressibility: $\nabla \cdot \overline{\boldsymbol{u}} = 0.$

CORRELATION TENSOR VERSUS TURBULENT STRESS

The term $-\nabla \cdot \boldsymbol{R}$ acts like an additional forcing in the nondimensional mean equation, with the same differential form as a stress divergence.

- $R_{ij} = \overline{u_i' u_j'}$ is a velocity-correlation tensor.
- In dimensional form, the corresponding turbulent (Reynolds) stress is commonly defined as $\tau_{ij}^R = -\rho R_{ij}$; some texts instead call ρR_{ij} the Reynolds-stress tensor and keep the minus sign in the equation.
- These are not determined by the mean equation alone.

Closure problem (core of RANS modeling)

The averaged equations introduce new unknowns ($\boldsymbol{R}$) that require modeling:

$$\boldsymbol{R} \approx \mathcal{M}(\overline{\boldsymbol{u}}, \nabla \overline{\boldsymbol{u}}, \ldots).$$

WHAT COMES NEXT (CLOSURE + ML)

Next topics in a RANS sequence:

- classic closures (e.g., eddy viscosity / Boussinesq hypothesis),
- k–ε, k–ω, Spalart–Allmaras, etc.
- data-driven closure ideas (ML approximations of $\boldsymbol{R}$ or corrections).

Takeaway

RANS reduces space-time complexity, but pays a price: **closure modeling**.

WORKED CHECK: FULLY DEVELOPED CHANNEL FLOW

Take a statistically stationary, fully developed channel, $\overline{u} = \overline{u}(y)$, $\overline{v} = \overline{w} = 0$, with statistics independent of x and z.

- The left side $\overline{u}\,\partial_x \overline{u} + \overline{v}\,\partial_y \overline{u} + \overline{w}\,\partial_z \overline{u}$ vanishes term by term.
- Of the Reynolds-stress terms only $\partial_y \overline{u'v'}$ survives, so the final x-equation becomes

$$0 = -\partial_x \overline{p} + \frac{1}{\mathrm{Re}} \frac{\mathrm{d}^2 \overline{u}}{\mathrm{d}y^2} - \frac{\mathrm{d}}{\mathrm{d}y} \overline{u'v'}.$$

- The y-equation gives $\overline{p} + \overline{v'^2}$ independent of y, so $\partial_x \overline{p}$ is a constant and the *total* shear stress $\frac{1}{\mathrm{Re}} \frac{\mathrm{d}\overline{u}}{\mathrm{d}y} - \overline{u'v'}$ is linear in y.

Interpretation

Near the wall the viscous part carries the stress; away from it the Reynolds shear stress $-\overline{u'v'}$ does. This is the starting point of the wall-scaling and log-law slides in deck 5-3 (see also Pope 2000, §7.1).

SUMMARY

- Start with x-momentum component of incompressible Navier–Stokes.
- Substitute $\boldsymbol{u} = \overline{\boldsymbol{u}} + \boldsymbol{u}'$ and time-average.
- Mixed mean–fluctuation terms vanish; products of fluctuations remain.
- Use product-rule identities + $\nabla \cdot \boldsymbol{u}' = 0$ to package surviving terms as

$$-\left(\frac{\partial}{\partial x}\overline{(u')^2} + \frac{\partial}{\partial y}\overline{u'v'} + \frac{\partial}{\partial z}\overline{u'w'}\right) = -(\nabla \cdot \boldsymbol{R})_x.$$

- The resulting mean equation requires a closure model for $\boldsymbol{R}$.

REFERENCES AND SIGN CONVENTION

- O. Reynolds, “On the dynamical theory of incompressible viscous fluids and the determination of the criterion,” *Philosophical Transactions of the Royal Society A*, 186 (1895). Pages 123–164; doi:10.1098/rsta.1895.0004.
- S. B. Pope, *Turbulent Flows*, Cambridge University Press (2000), Chapters 3–4.
- S. L. Brunton, lecture on Reynolds-averaged Navier–Stokes equations. YouTube (the video linked from Part 1; the derivation sequence of this deck is adapted from Brunton’s RANS lecture material).
- D. C. Wilcox, *Turbulence Modeling for CFD*, 3rd ed., DCW Industries (2006), Ch. 2 (Reynolds averaging and the Reynolds-stress tensor; alternative sign conventions).

Convention used here

$R_{ij} = \overline{u_i' u_j'}$ is the velocity-correlation tensor, so the nondimensional RANS equation contains $-\partial_j R_{ij}$. In dimensional stress notation, $\tau_{ij}^R = -\rho R_{ij}$ and the same contribution is $\rho^{-1}\partial_j \tau_{ij}^R$.